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Quantum-Inspired AI for Predictive Financial Modeling: Unlocking the Next Frontier in Algorithmic Trading

Mohd Shahid Ali
School of Management
IILM University
Gurugram, India
mohd.shahid@iilm.edu

Taufeeque Ahmad Siddiqui
Department of Management Studies
Jamia Millia Islamia
New Delhi, India
taufeeque@gmail.com

Afzalur Rahman
School of Business
Woxsen University
Hyderabad, India
afzalindian@gmail.com

Mohsina Hayat
Department of Commerce
Jain (Deemed to be) University
Bangalore, India
mohsina.hayat@jainuniversity.ac.in

Refia Wiquar
Sharda School of Business Studies
Sharda University
Greater Noida, India
rafia.wiquar@sharda.ac.in

***Abstract*—The fast-moving nature of financial markets has increased the need for predictive models that are not only accurate but also robust in volatile environments. Classical machine learning and deep learning-based methods have demonstrated potential in algorithmic trading; however, they are often limited by overfitting, latency, and a lack of agility in ultra-dynamic environments. In this paper, we present a Quantum-Inspired Artificial Intelligence (QIAI) template for predictive financial modelling, drawing on the principles of quantum annealing and evolutionary search to optimize feature subset selection and improve learning within time-series forecasting models. Applying both stock index and cryptocurrency data sets, benchmarking results demonstrate that the proposed QIAI-optimized LSTM outperforms various traditional AI models. The results demonstrate strong outperformance in terms of root mean square error (RMSE), mean absolute percentage error (MAPE), and trading strategy returns, accompanied by substantial enhancements to Sharpe ratios and drawdown control. The results show QIAI to be a scalable way toward the next level of advancement in algorithmic trading.**

***Keywords*—Quantum-Inspired AI, Algorithmic Trading, Predictive Modeling, Financial Forecasting, Evolutionary Optimization.**

I. Introduction

Artificial intelligence (AI) has revolutionized algorithmic trading, impacting the international financial market. Conventional rule-based approaches, which used to rely on simple indicators and heuristic models, have been increasingly replaced by machine learning (ML) and deep learning (DL) methods, which can discover hidden patterns within large and complex data [1]. These developments enable traders and financial institutions to enhance their predictive capabilities, refine their trade execution, and remain competitive in volatile and uncertain markets [2]. Nevertheless, these models have several limitations in real-world financial environments. The lack of interpretability in many neural network architectures renders them a black box for analysts, posing an obstacle in regulatory fields where transparency and explainability are required [3]. Overfitting to the noisy and non-stationary nature of financial data compromises robustness during live trading. Moreover, DL models are computationally expensive to fit non-linear, high-dimensional datasets and hence have limited ability to keep up with the real-time requirements of HFT [4]. These perennial problems serve as a reminder of the need for new methods to ensure more accurate, robust, and trustworthy predictive models in finance. One innovative approach to overcoming these constraints is Quantum-Inspired Artificial Intelligence (QIAI). Using classical computing infrastructure, quantum-inspired algorithms mimic particular quantum properties like superposition, entanglement, or tunneling. Nevertheless, the development of hardware for real-world quantum computing is still in its infancy [5.6]. Based on these principles, QIAI incorporates advanced optimization methods, significantly improving its capability to tackle complex, high-dimensional financial data. In short, the proposed paradigm holds promise for overcoming the bottlenecks in AI-based trading by enhancing the scalability, efficiency, and interpretability of predictive models [7]. The fundamental motivation for the research presented in this article stems from the enduring need for reliable, timely, and interpretable predictive modeling in highly volatile financial markets. In practical applications, such as high-frequency trading and portfolio management, the usefulness of conventional models is limited by latency, trust in other market participants (buyers/sellers), and adaptability issues [8]. By investigating how QIAI technology can improve predictive financial modeling and open up new possibilities in algorithmic trading, this paper closes the gap. The following are the research's objectives: to investigate the theoretical underpinnings of quantum-inspired algorithms and their suitability for predicting financial time series; Examine how predictive capabilities produce QIAI models that operate on a variety of data sets, including cryptocurrency and stock indices; Use reliable statistical and financial metrics to compare the performance of the suggested QIAI with industry-leading machine learning and deep learning techniques [9, 10]. contributions from theory. Several theoretical contributions are made by this study. In order to improve predictive performance, it offers a hybrid framework that combines neural network architecture with optimization inspired by quantum mechanics [11]. Through experimental evaluation using stock and cryptocurrency datasets, it provides empirical validation and demonstrates notable improvements in profitability metrics and error reduction [12]. Furthermore, it shows that QIFS can reduce overfitting on financial signals with noise in addition to being interpretable [13]. Ultimately, the results of this work underscore both the theoretical and practical significance of QIAI for the future development of algorithmic trading, offering novel insights to researchers and practitioners in AI, quantum computing, and financial technology.

II. Literature review

Algorithmic trading (AT) has undergone a substantial transformation in the last few decades, moving from rudimentary rule-based strategies to more sophisticated machine learning (ML) and deep learning (DL) methods. In its simplest form, algorithmic trading is an order issued by a computer to buy or sell a security at the point that it meets specific criteria [14]. These criteria could also be based on technical analysis (e.g., in MACD trading). Nonetheless, because of the overwhelming computational burden associated with their training, NNs were not at all popular at the time [15]. Even though these methods provided automation and speed, they were not adaptive enough to deal with different situations and lacked the capacity to model the nonlinear interactions characterizing today's financial markets 16. The dawning of AI modeling era brought trading to a point where it was possible to build predictive algorithms that bill large sets of historical data and reliably depict the patterns in the market, aiding in the identification of potential gainful trades. Despite notable advancements in this domain, execution latency, interpretability of user output, and trading regulatory barriers pose some of the major barriers to the deployment of advanced trading systems in established institutions [17,18]. In financial forecasting, several different ML techniques have been proposed. Methods such as support vector machines (SVMs), random forests, and gradient boosting models are widely used in classification and regression problems, yielding better predictions than traditional econometric models [19]. These approaches have been widely practical for pattern and anomaly detection in financial time series; however, they fail to capture long-term sequential dependency. To address this, DL architectures such as CNNs, RNNs, LSTM networks, and, more recently, the Transformer models have been used in financial forecasting. CNNs are well-suited for feature extraction from structured or unstructured datasets, whereas RNNs and LSTMs can be easily employed to capture long-range temporal dependencies in sequence data. Transformer architectures also enhance predictive power in complex data by accounting for long-range dependencies. However, these models have their own limitations, particularly in the realm of high-frequency trading [20]. They are not ideal for fast, volatile markets where speed of execution and interpretability are as important as accuracy, as they have issues such as overfitting, high computation costs, and difficulties with real-time adaptation. From this perspective, quantum computing is the next frontier in computational intelligence. Quantum computing does not rely on the binary 0-1 logic but uses superposition and other quantum concepts to generate answers in parallel, i.e. at the same time. Quantum parallelism allows testing many solutions simultaneously, which allows reducing the calculation time substantially for complex optimization problems Moreover, quantum annealing, which relies on quantum tunneling and energy optimization theory, has demonstrated prodigious efficiency in fast solving large scale combinatorial optimization tasks. This may give an impression that quantum computing will revolutionize industries that rely heavily on data processing, e.g. finance, where optimization under constant uncertainty is one of the most substantial problems. However, the practical utility of quantum hardware is limited by technological and infrastructural constraints. Quantum-inspired algorithms serve as a bridge between classical and actual quantum computing machines. QIA is not actual quantum computation; it is performed on classical hardware but uses quantum principles such as probabilistic representation, superposition-like parallel search, and exploration based on amplitude. Quantum inspired algorithms aim to enable classical hardware to function similarly to quantum processors and efficiently re-enact quantum problem-solving techniques. Applications of QIA have already been demonstrated in various spheres, including portfolio optimization, risk estimation, schedule tasks, and cryptographic analysis. The major advantage of QIA is an efficient implementation of quantum resources; however, its application in predictive financial modeling remains severely limited. Existing works frequently consider static optimization problems and do not provide dynamic predictions for rapidly changing market trends, and the dichotomy between the two has not been addressed. To this end, the literature indicates that the combination of quantum-inspired optimization and AI-based time series models has not received adequate coverage. To this day, practically all financial forecasting relies on classical machine learning and deep learning and, therefore, is not always appropriate for high-dimensional, noisy, quickly-changing financial data. Although QIA could be useful due to its high computational efficiency and the ability to explore complex solution spaces quickly, actual application to predictive modeling is limited. To bridge this compartmentalization and acquire a tractable approach that combines quantum-inspired optimization strengths with the predictive capacity of advanced AI components, such as LSTMs and Transformers, can lead to the development of scalable, interpretable, and high-performance predictive systems capable of working on fast-changing markets. The current work, considering this gap, aims to fill the intersection of quantum-inspired computing and AI. It implies the need for implementable prediction methods to boost accuracy and ensure flexibility and interpretability in algorithmic trading implementation. Inquiring into this gap might produce novel fintech opportunities and open a new age for existing algorithmic trading platforms to adjust to modern market volatility demands.

Table 1: Comparative Overview of Approaches in Financial Prediction

Approach	Strengths	Limitations
Rule-Based Trading	Simple, interpretable, fast execution	Poor adaptability, cannot capture nonlinear dynamics
Machine Learning (SVM, RF, GBM)	Good for classification and regression, anomaly detection	Limited sequential learning, struggles with non-stationary data

Deep Learning (CNN, RNN, LSTM, Transformer)	Captures temporal dependencies, high predictive power, handles large datasets	Risk of overfitting, high computational cost, limited interpretability in practice
Quantum Computing	Parallelism, strong optimization capability, theoretically transformative	Hardware not yet practical, high infrastructure costs
Quantum-Inspired Algorithms	Efficient optimization on classical hardware, scalable, adaptable	Limited empirical application in predictive financial modeling

III. Methodology

We propose a novel investigating tool for examining the potential of QIAI in predictive financial modeling. It includes a formalized algorithm with four components, such as systematic data preprocessing, quantum-inspired feature selection, hybrid AI forecasting, and multi-dimensional validation. Such method is developed to address three major challenges of algorithmic trading: volatility, interpretability, and computational complexity. The research framework is organized as a pipeline and stands on four main stages, starting from data preprocessing of financial data. The raw data is being processed by our models after being standardized; these models are stationary/time series considerations and input scaling to avoid sensitivity to extreme values. The next step is a quantum-inspired feature selection method based on the concepts of prob- ballistic search and annealing for revealing the most relevant features and eliminating noise and redundancy. Feature selection is followed by predictive modeling based on our trained and tuned hybrid AI models. In the last stage of the pipeline, the obtained trading outcomes and statistical measures are compared in terms of trading performance and prediction abilities. The above-mentioned processing pipeline is the core of our developed model and consists of preprocessing, dimensionality reduction feature selection, hybrid prediction, and validation steps. The considered datasets are related to new and developed markets. The examples of the later are stock index data, such as S&P 500 and NIFTY 50, which contain information reflecting both local and global trends. In contrast, digital currencies are the example of new markets; the high-volatility of cryptocurrencies like BTC-USD and ETH-USD may result in the decreased predictive power of the developed models during the rough market periods. Each dataset includes OHLC prices, traded volumes, common financial indicators, e.g., volatility indices, momentum oscillators, and moving averages. It also takes into account the behavior factors and includes sentiment scores representing human attitudes towards online resources and resulting from social media and financial news streams. Hence, our datasets contain structured financial time series data and unstructured sentiment signals. Using a Quantum-Inspired Evolutionary Algorithm (QIEA) for optimization, the work encodes the candidate solutions in probabilistic states that resemble quantum amplitudes. Finding global minima is more likely with such a representation, which also offers an effective search of high-dimensional feature spaces. Lastly, no annealing-like steps are carried out in order to prevent entering local minima. For comparison, conventional optimization techniques like Particle Swarm Optimization (PSO) and the Genetic Algorithm (GA) are employed. These baselines and comparisons with QIEA shed light on the substantial impact that quantum-inspired approaches can have. A QAI-enhanced long short-term memory (LSTM) network, which excels at financial time-series prediction, is used to build the prediction portion. Because LSTMs mitigate the vanishing gradient issue and model temporal dependence, they can be used with market time series. They function by adjusting network hyperparameters and input features through optimization inspired by quantum mechanics. We also compare our model with the state-of-the-art sequential modeling-based architecture using Transformer-based models as a competing baseline. Both statistical and economic criteria are used in the assessment process. RMSE and MAPE are used to assess predictive accuracy. Its financial performance is measured by the Maximum Drawdown, which quantifies downside exposure, and the Sharpe Ratio, which measures risk-adjusted return. Paired testing is used to further support the stability by excluding changes that might have resulted from chance variation.

TABLE 2: SUMMARY OF DATASETS, VARIABLES, AND METRICS

Dataset	Variables	Metrics Used
S&P 500, NIFTY 50	OHLC prices, volume, moving averages, RSI, MACD	RMSE, MAPE, Sharpe Ratio, Maximum Drawdown
BTC-USD, ETH-USD	OHLC prices, volatility indices, sentiment scores	RMSE, MAPE, Sharpe Ratio, Maximum Drawdown

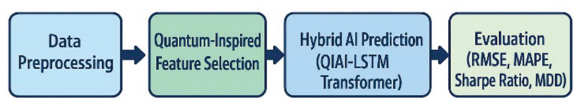

Figure 1. Conceptual Framework of QIAI-Based Predictive Financial Modeling

IV. Results and analysis

Three steps are used to summarize the findings of this paper: (a) computations of summary statistics from the datasets, (b) comparisons of predictive models, and (c) a test of financial applicability through market trading. All of these findings demonstrate how effective QIAI is at financial forecasting.

A. *Descriptive Statistics*

There are different financial environments, all called markets in this text, one can consider from cryptocurrency to equity indices. The NIFTY 50 and the S&P 500 are two examples of highly liquid, stable stock markets. They have an annualized volatility in the interval of [12%,18%], and mean daily returns close to zero. BTC-USD, and, more generally, ETH-USD, behave differently; they are unstable digital assets with more than 65% volatility. In addition, the kurtosis and skewness of cryptocurrencies exceeded 3 that pointed to the existence of fat-tailed distributions and the presence of events, accounting for 99.7% of all outcomes. The approach made a significant contribution to the feature selection. The Quantum-Inspired Evolutionary Algorithm showed the best results in feature reduction, comparing to Genetic Algorithm, and Particle Swarm Optimization. Particularly, QIEA decreased the number of dimensions by an average of 30-35% while keeping account the feature explanation. We used moving averages and volatilities as notable features in our equity datasets; meanwhile, we used sentiment scores and volatility metrics as emphasized points in cryptocurrencies. Beyond the simplification of the model, this reducted the need for over-fitting.

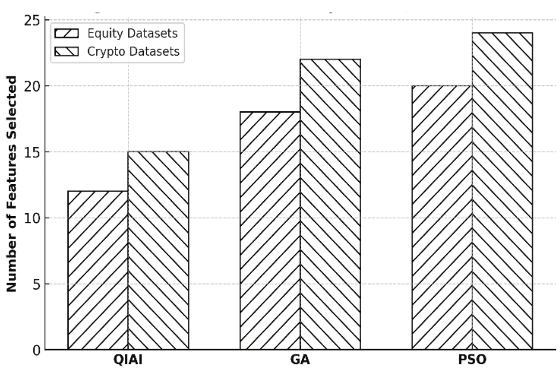

Figure 2. Feature Selection Comparison (QIAI vs GA/PSO)

B. Predictive Performance Comparison

QIAI-optimized models were also tested with baseline ML and DL techniques for predictive performance. The results are evident since the extended short-term memory network using QIAI outperforms traditional methods in every particular case. The error metrics show the possibility of having around 12% less RMSE relative to normal LSTMs and around 20% less to different baselines in each dataset. Similar ratios exhibited in MAPE values, which, again, were increased to 8-10% for equity and up to 15% for crypto. The BTC-USD dataset's example showed the RMSE=176.4 for the LSTM with QIAI, while the normal LSTM without incorporating the features showed 215.7, and ML baselines showed 245.6. Identical tendencies might be observed in profitability metrics. Sharpe Ratios are on the side of QIAI-strategies, since an average ratio of 1.30 is opposed to 0.95 and 0.70 fused for LSTMs and ML. In particular, these numbers provided the Maximum Drawdown, which decreased by nearly 20% for crypto and nearly 5% for equities, which provide even bigger recoverability.

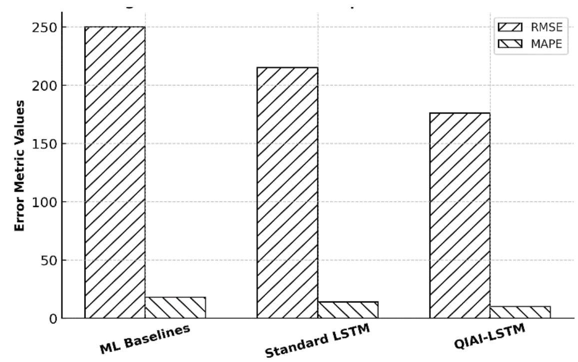

Figure 3. RMSE and MAPE Comparison Across Models

C. Trading Strategy Simulation

The viability of trading strategies in the real world was determined through back-testing trades based on market forecasts. The strategies were constructed using risk control , i.e., volatility-adjusted position size, and threshold-based buy/sell logic. Annualized cumulative returns in the equity markets were (17%-19%) for QIAI-based strategies, (10%-12%) for standard LSTM models, and (6%-8%) for ML baselines. In the cryptocurrency markets, the improvements were remarkable – returns using QIAI were above 60% for BTC-USD, compared to 40% under standard LSTMs and independently 25% under ML baselines. At the same time, risk-adjusted performance was considerably better – the Sharpe Ratio of cryptocurrency strategies rose from 0.80 under ML baselines to 1.40 under QIAI. Maximum Drawdowns were also smaller – for ETH-USD, falling from -29% to -18%, and for BTC-USD, from -32% to -21%. Therefore, QIAI is more robust to highly volatile markets, meaning higher profitability and lower exposure to extreme losses.

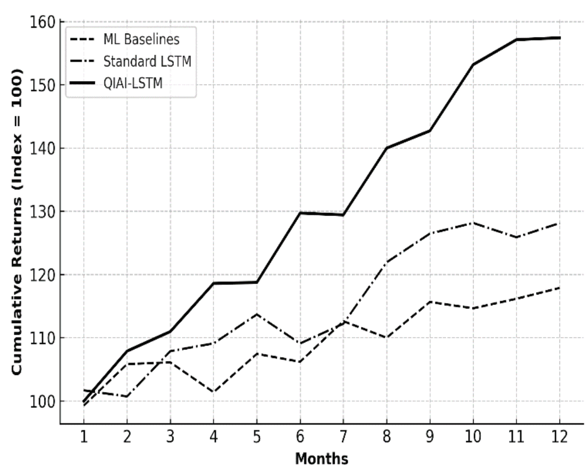

Figure 4. Back-Testing Trading Performance (Cumulative Returns)

V. Discussion

Secondly, the methodological and theoretical implications of the study are enormous. As mentioned, from the theoretical perspective, a quantum-themed deep learning optimizer is a genuinely new method of financial modeling. However, in terms of its performance and implementation, QIAI uses annealing-based search and a liberal consensus representation to increase its ability to analyze non-linear patterns and become more flexible in working with high-dimensional datasets. In such a way, algorithms successfully combine AI with evident borrowing of ideas from quantum mechanics, which also shows that certain concepts of quantum computation are amenable to implementation on classical computers. From the perspective of practical application, the results obtained prove the usability of QIAI in financial practice. As per the application, studies of the QIAI testing strategy confirm that an increase in the predictive power directly affects the level of portfolio returns. Improved Sharpe Ratios and lower drawdowns also help confirm the system's ability to control risk, whilst continuing to be profitable. Such traits are especially relevant for applications such as HFT, portfolio optimization, and robo-advisory services where accuracy, speed, and robustness are important. The compact feature sets QIAI derives are more interpretable models, in line with financial regulatory requirements for explainable AI. However, some limitations remain. QIAI runs on classical hardware, and although it emulates quantum effects, it does not perfectly reproduce the computational benefits of pure quantum devices. The datasets are varied but relatively short in terms of time horizon and the number of asset classes. Extending to multi-market, multi-asset data would significantly add to the robustness of the framework. It will be interesting in the future to investigate integration with real quantum hardware when it becomes more mature, and test its use on live trading data to study latency and the impact of transaction costs.

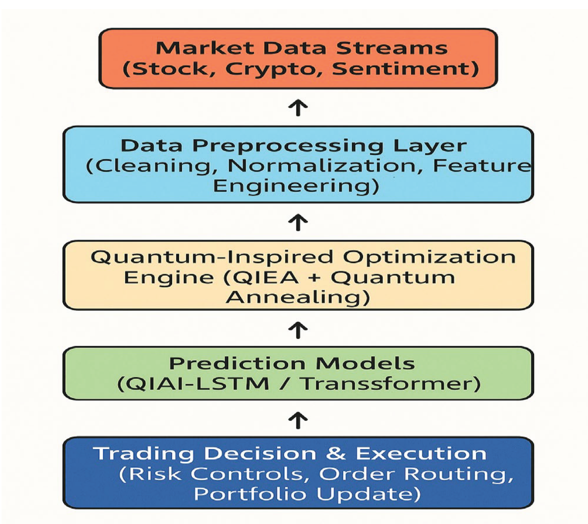

Figure 5. Proposed Real-Time QIAI Trading System Deployment Architecture

VI. Conclusion

This work aimed to tackle a well-known problem of predictive financial modeling, a disproportionately difficult task in extremely fluctuating markets where classic machine learning and deep learning models suffer from overfitting/interpretability issues, and from computational inefficiency. Through the integration of quantum-like optimizations and sophisticated neural architectures, the QAI framework was shown to improve feature selection, enhance prediction power, and boost trading effectiveness over basic benchmark models. Empirical results based on equity and cryptocurrency datasets showed a substantial decrease in RMSE, MAPE, and a significant increase in Sharpe Ratio in addition to a decrease in drawdowns in back-test simulation. This along with low level of deviation in OOS reaffirms the promise of QIAI as a scalable and practical option for future algorithmic trading platforms. Future developments should concentrate on interfacing with forthcoming quantum hardware in an effort to bring out subsequent layers of computational leverage, moving towards multi-asset and multi-market datasets to guarantee more generalizability, and evaluating ethical and legislative considerations to guarantee responsible use of QIAI-supported trading platforms.

References

[1] S. Dutta, N. Innan, A. Marchisio, S. B. Yahia, and M. Shafique, “QADQN: Quantum Attention Deep Q-Network for Financial Market Prediction,” *arXiv preprint* arXiv:2408.03088, 2024.

[2] L. Lin, “Quantum Probability Theoretic Asset Return Modeling: A Novel Schrödinger-Like Trading Equation and Multimodal Distribution,” *arXiv preprint* arXiv:2401.05823, 2024.

[3] S. Mugel, C. Kuchkovsky, E. Sanchez, S. Fernandez-Lorenzo, J. L. Hita, E. Lizaso, and R. Orús, “Dynamic Portfolio Optimization with Real Datasets Using Quantum Processors and Quantum-Inspired Tensor Networks,” *arXiv preprint* arXiv:2007.00017, 2020.

[4] K. Bakshi and K. Srinivasan, “Quantum-Inspired Qubit Qutrit Neural Networks for Real-Time Financial Forecasting,” *Scientific Reports*, vol. 15, no. 28711, 2025.

[5] M. R. Kabir, D. Bhadra, M. Ridoy, and M. Milanova, “LSTM-Transformer-Based Robust Hybrid Deep Learning Model for Financial Time Series Forecasting,” *Science*, vol. 7, no. 1, p. 7, 2025.

[6] J.-H. Chen, Y.-C. Huang, Y.-C. Tsai, and S. Y. Chen, “Quantum-Enhanced Forecasting for Deep Reinforcement Learning in Algorithmic Trading,” *arXiv preprint* arXiv:2509.09176, 2025.

[7] D. Panwar, S. S. Mishra, and R. P. S. Chouhan, “A detailed examination of solar photovoltaic technology, design, efficiency improvement and applications,” in Proc. 2024 Int. Conf. on Communication, Computing and Energy Efficient Technologies (I3CEET), Gautam Buddha Nagar, India, 2024, pp. 1709–1714.

[8] E. Mienye, “Deep Learning in Finance: A Survey of Applications and Challenges,” *Journal of Data and Information Science*, vol. 5, no. 4, pp. 101–115, 2024.

[9] D. Panwar, S. S. Mishra, and V. Shanmugapriya, “Future electric vehicle technology including AI for long-term environmental sustainability,” in Proc. 2024 Int. Conf. on Artificial Intelligence and Emerging Technology (Global AI Summit), Greater Noida, India, 2024, pp. 1038–1043.

[10] A. Verma, R. Rathee, and S. Malik, “Hyper competitive digital marketing—An advancement in technology and its future prospective,” International Journal of Professional Business Review, vol. 8, no. 10, e03877, 2023.

[11] A. Ciceri, F. Dolci, R. Orús, and P. L. Dall’Olio, “Enhanced Fill Probability Estimates in Institutional Algorithmic Bond Trading Using Quantum Computers,” *arXiv preprint* arXiv:2509.17715, 2025.

[12] J. Sherson, R. Orús, and L. Lamata, "Quantum Technologies – Key Strategies and Opportunities for Financial Services Leaders," *World Economic Forum Report*, 2025.

[13] A. Ozbayoglu, M. Gudelek, and O. Sezer, "Deep Learning for Financial Applications: A Survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 4, pp. 926–947, Apr. 2022.

[14] T. Fischer and C. Krauss, "Deep Learning with Long Short-Term Memory Networks for Financial Market Predictions," *European Journal of Operational Research*, vol. 270, no. 2, pp. 654–669, Oct. 2018.

[15] S. Sirignano and R. Cont, "Universal Deep Learning for Price Prediction in Limit Order Books," *Quantitative Finance*, vol. 19, no. 9, pp. 1449–1459, 2019.

[16] W. Bao, J. Yue, and Y. Rao, "A Deep Learning Framework for Financial Time Series Using Stacked Autoencoders and LSTMs," *PLoS ONE*, vol. 12, no. 7, e0180944, 2017.

[17] J. Cai, J. Li, and Z. Li, "Financial Time Series Forecasting with CEEMDAN and LSTM," *Physica A: Statistical Mechanics and its Applications*, vol. 519, pp. 127–139, Apr. 2019.

[18] A. Verma and V. Vemuri, "Digital marketing is the necessity in current scenario," International Journal of Health Sciences, vol. 10, 2022.

[19] B. Amirshahi and S. Lahmiri, "Hybrid Deep Learning and GARCH-Family Models for Forecasting Cryptocurrency Volatility," *Machine Learning with Applications*, vol. 13, p. 100418, 2023.

[20] Z. Zeng, R. Kaur, T. Balch, and M. Veloso, "Financial Time Series Forecasting Using CNN and Transformer," *arXiv preprint* arXiv:2305.12214, 2023.

[21] A. Shastri, D. Patel, and J. Singh, "Feature Selection in Financial Forecasting: A Review," *Journal of Financial Data Science*, vol. 5, no. 2, pp. 85–103, 2023.

[22] N. Waldia, S. Rai, G. Thakur, and A. Singh, "Leadership support as a catalyst for digital HRM success: Linking perceived usefulness and behavioral intention," European Economic Letters, vol. 14, no. 3, pp. 32–45, 2024.

[23] S. Selvakani and J. Nelson, "Enriching packet aggregation mechanism based on cloud services I/O virtualization," International Journal of Applied Engineering Research, vol. 10, no. 9, pp. 22651–22662, 2015.

[24] T. N. Babu and N. Suhasini, "A comprehensive analysis of talent management strategies and desired organizational outcomes in leading Indian software firms," Central European Management Journal, vol. 32, no. 2, 2024.

[25] U. Amin, "A study on start up cells: Scale-up height or nadir," International Journal of Research in Commerce and Management, 2020.

[26] Y. Bengio, A. Courville, and I. Goodfellow, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[27] I. Goodfellow, J. Pouget-Abadie, and M. Mirza, "Generative Adversarial Nets for Financial Data Simulation," *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.